

**Physical and Perceptual Costs Trade Off in Decision-Making: Based on a
Target-Hitting Game**

PSYCH 505, DIS 307, Group 4

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Introduction (Written by: Paige)

On a daily basis, we encounter decisions in which multiple options offer the same reward, but we must select one. Beyond the reward (what we gain) itself, people tend to consider how simple or complicated each option feels, including physical, mental, and perceptual demands. In these cases, people tend to prefer options that require less effort in general. This human tendency aligns with current literature showing that individuals routinely avoid the more cognitively demanding choice if possible (even when rewards are similar) (Kool et al., 2010).

This human tendency is mirrored in modern models of decision-making, which indicates that the brain evaluates a tradeoff between both rewards and costs when guiding our behavior. For instance, new models of decision-making imply that the brain evaluates and weighs negative factors associated with committing to an action, such as physical effort, time, or cognitive demand (Rushworth & Behrens, 2008). These aspects are collectively referred to as action cost. At the same time, the expected advantages of each option are considered separately and termed as stimulus value. Together, action cost and stimulus value are combined into an overall term called action value, which ultimately guides behavior (Rangel & Clithero, 2012).

Furthermore, this framework is also supported by neuroscience research, which demonstrates that distinct brain regions are involved in processing these components. For example, neural evidence shows that brain regions, including the striatum, encode action-specific reward values that directly guide decision-making behavior (Samejima et al., 2005). The prefrontal cortex has been shown to encode reward-related cues, which play a significant role in directing reward-seeking behavior (Otis et al., 2017). On the other hand, the ventromedial prefrontal cortex (vmPFC) and orbitofrontal cortex (OFC) are commonly associated with representing stimulus value, while the anterior cingulate cortex (ACC) plays a critical role in evaluating action cost (Rushworth & Behrens, 2008).

At the same time, action cost is not limited to only physical effort, but also includes perceptual and cognitive demands, such as the difficulty of visually processing or identifying a target. During this type of perceptual decision-making, the brain seems to rely on evidence accumulation processes, where the brain integrates sensory information over time before officially committing to a decision (Shadlen & Kiani, 2013).

This reveals that decision-making reflects a tradeoff between multiple forms of cost (even when the reward remains constant). However, it is still unclear how individuals evaluate different types of costs against each other (for example, when physical and perceptual costs compete). In this context, physical effort can be represented by movement distance, while perceptual cost can be represented by how visually clear (salient) or easy a target is to identify/process.

To address the current gap in research, this study examines how participants make decisions when physical distance and perceptual salience are manipulated across different conditions. For example, we compare behavior when we align conditions (where both factors influence the same option), manipulate distance and salience separately (where one variable changes), and conflict trial conditions (where distance and salience favor different options).

We hypothesize that when reward is held constant, participants will choose targets that minimize action cost. More specifically, targets that are closer to the cursor (lower physical cost) and more visually salient (lower perceptual cost) will be chosen more frequently. Additionally, when these factors conflict, we expect that physical cost (distance) will have a stronger influence on choice over perceptual cost. This expectation correlates with our first research question (RQ1), which examines how choice behavior varies across these conditions to determine whether participants prioritize movement cost, perceptual reliability, or balance these two options. Moreover, we further explore this concept in the second research question (RQ2), which tests whether feedback from participants' previous trial (like large errors) influences subsequent choices (specifically if there is a shift toward more reliable alternatives after high error trials). Ultimately, we want to further know if any results we find are generalizable to population level or individual differences.

Methods (Written by: Jenna; edited by Paige)

Participants

We recruited 36 participants aged 18 years or older to participate in the study. Participants were recruited primarily by sharing the task with family members and friends. Participants were asked to provide demographic information, including age,

gender, and handedness at the beginning of the task. The mean age was 21.83 years (SD = 3.06, range = 18–34). The sample included 21 females (58.3%), 14 males (38.9%), and 1 non-binary participant (2.8%). All participants were right-handed. After data quality screening, 7 participants were excluded, leaving a final sample of 29 participants for analysis.

Stimuli

Targets. The stimuli were two circular targets presented on the left and right sides of the screen. Each target was 180 px in diameter.

Distance. Target distance was manipulated using three possible distances from the center of the screen: close (180 px), medium (320 px), and far (460 px).

Salience. Visual salience was manipulated using blur filters. There were three salience levels: high blur (0px), medium blur (8px), and no blur (16px).

Procedure

The experiment was conducted as a web-based task with the code managed in GitHub (assisted by Gemini and ChatGPT). Participants first completed 4 practice trials to become familiar with the task, followed by 40 main experimental trials.

At the beginning of each trial, a circular button appeared at the center of the screen. Participants were instructed to press and hold the button for 500 ms and maintain visual fixation on it. After the hold period, the button disappeared and two targets appeared on the screen.

Participants were instructed to click as accurately as possible at the center of one of the targets. After each click, feedback appeared at the center of the screen for 1000 ms, showing the distance (in pixels) between the click location and the center of the selected target and whether the response was a "Perfect" (0 px), "Hit" (0-5 px), "Almost there" (5-20 px), or "Miss" (> 20 px). Feedback was presented for 1000 ms before the next trial began.

Data Collection

Trial-level data were sent to an online Google sheet after the experiment ended. For each trial, their choice (left/right), reaction time, the click location, and error (distance from center) are recorded.

Data Analysis

We used RStudio (2025.05.1+513) for statistical analysis (assisted by chatGPT). Practice trials were excluded before the main analyses.

Participants were excluded for poor accuracy (main-trial accuracy within 20 px < .80) or extreme reaction time (median reaction time > 5000 ms or more than 20% of trials with reaction times > 8000 ms). After the screening, we excluded 7 participants (originally 36 participants), so a final sample of 29 participants was included in the analysis.

For the analysis, we first tested whether participants chose the optimal target above chance using binomial tests. We then use one-sample t-tests for participant-level choice rates. The relationship between previous-trial error and subsequent choice was analyzed using correlation and logistic regression. Changes in behavior over time were examined using logistic regression.

Results (Written by: Jenna, Paige; edited by Patrick)

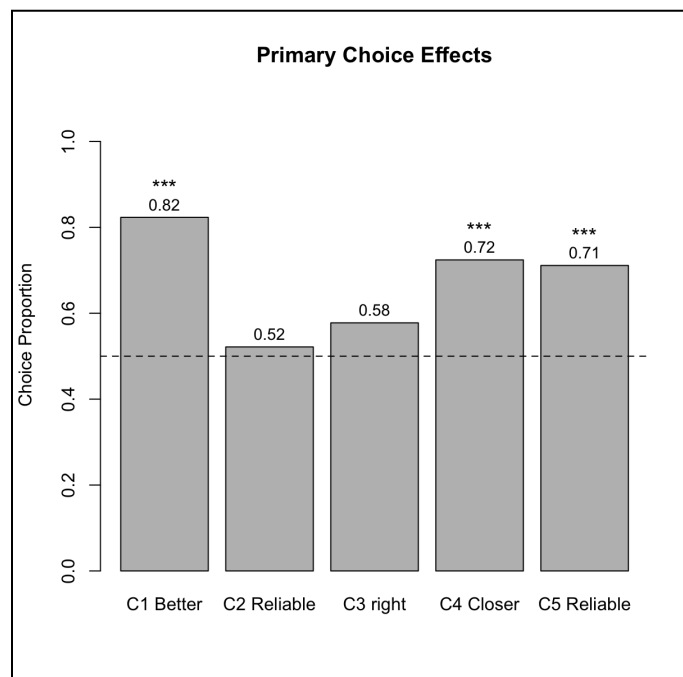


Figure 1. (RQ1) Choice behavior across different task conditions. The proportion of trials where participants selected the optimal target across congruent, distance-only, salience-only, conflict, and equal conditions. The dashed line is the chance level (0.5). Individuals showed stronger preferences for the optimal target in congruent,

distance-only, and salience-only conditions, but for conflict trials, the choices were around chance level, so it indicates a tradeoff between movement cost and perceptual reliability.

In congruent trials (Condition 1), participants chose the optimal target on 82.3% of trials, which was significantly above chance ($z = 9.85$, $p < .001$, $h = 0.70$).

In distance-only trials (Condition 4), participants chose the closer target on 72.4% of trials ($z = 6.83$, $p < .001$, $h = 0.47$). Similarly, in salience-only trials (Condition 5), participants chose the more reliable target on 71.1% of trials ($z = 6.43$, $p < .001$, $h = 0.44$).

In conflict trials (Condition 2), participants chose the more reliable target on 52.2% of trials, which did not differ from chance ($z = 0.66$, $p = .555$, $h = 0.04$).

In equal trials (Condition 3), participants showed a small but significant right-side bias, selecting the right target on 57.8% of trials ($z = 2.36$, $p = .021$, $h = 0.16$).

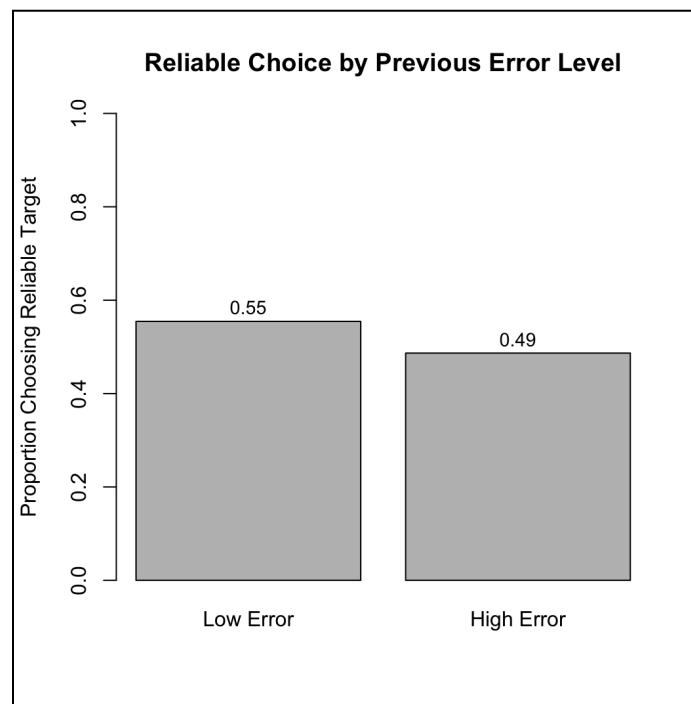


Figure 2. (RQ2) Influence on previous trial on subsequent choice. The proportion of trials where participants selected the more reliable target after low and high error trials (median split). No significant difference was observed between conditions, indicating that previous trial error did not strongly influence subsequent decisions.

Previous-trial error was split into low-error and high-error groups using the median as a cutoff, with values below or equal to the median classified as low error and values above the median classified as high error.

Participants chose the more reliable target on 55.5% of trials following low-error trials and 48.7% following high-error trials ($t(229.23) = 1.03$, $p = .303$, $d = -0.14$, 95% CI $[-0.062, 0.197]$).

Previous-trial error was not associated with subsequent reliable choice ($r = .021$, $t(230) = 0.31$, $p = .754$, 95% CI $[-0.108, 0.149]$).

A logistic regression analysis further showed that previous error did not predict reliable choice ($\beta = 0.005$, $SE = 0.017$, $z = 0.31$, $p = .753$, $OR = 1.01$, 95% CI $[0.97, 1.04]$).

Exploratory analysis comparing hit versus miss trials also showed no clear effect (51.6% vs. 66.7%).

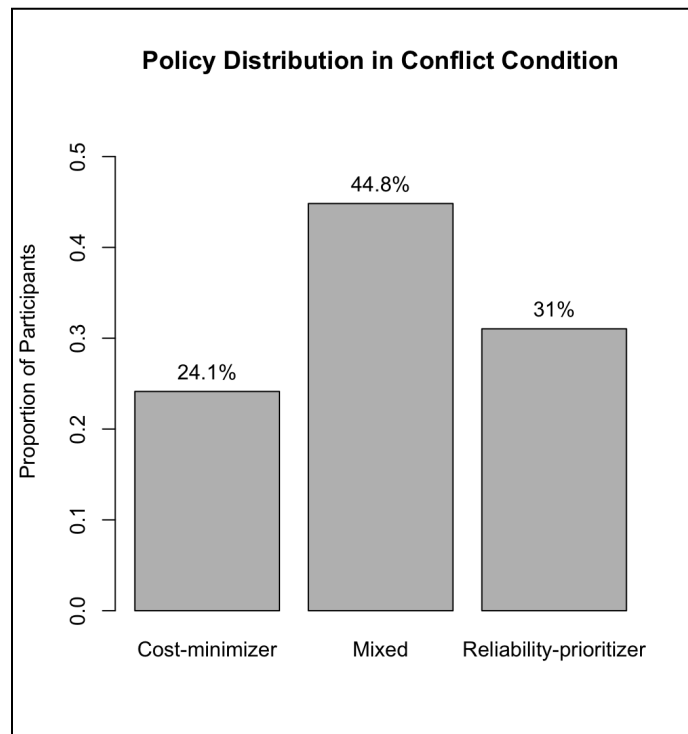


Figure 3. Distribution of decision-making across participants. Participants were categorized as cost-minimizers, mixed, or reliability-prioritizers (based on reliable

choices in conflict trials). The distribution shows variability in how individuals weigh movement cost versus perceptual reliability.

Participants were classified into three strategy groups based on their conflict-trial choices (Figure 3). Those who chose the reliable target on at least 65% of trials were classified as reliability-prioritizers, those who chose it on 35% or fewer trials were classified as cost-minimizers, and those in between were classified as mixed.

A total of 7 participants were classified as cost-minimizers, 13 as mixed, and 9 as reliability-prioritizers. The distribution of strategies did not differ significantly from equal proportions ($\chi^2(2) = 1.93$, $p = .381$, Cramer's $V = 0.18$).

At the participant level, the average proportion of choosing the reliable target in conflict trials was 52.2%, which did not differ from chance ($t(28) = 0.37$, $p = .714$, $d = 0.07$, 95% CI [0.402, 0.641]).

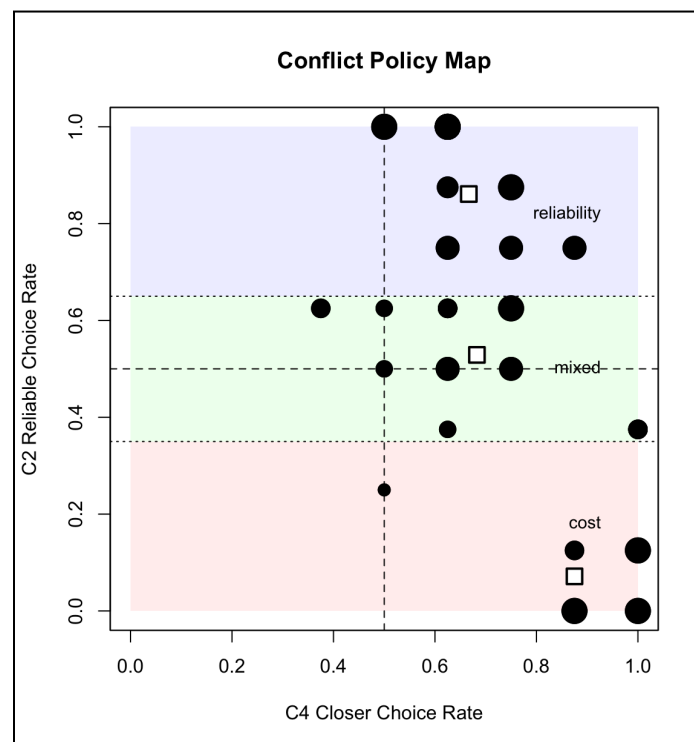


Figure 4. Conflict policy map showing decision-making strategies across conditions. Each point represents a participant, with reliable choices in conflict trials (y-axis) and closer choices in distance-only trials (x-axis). Point size reflects

salience-only choices (reliability), and shaded regions indicate strategy groups (cost-minimizer, mixed, reliability-prioritizer).

Participants previously classified as being part of a strategy group based on the conflict-trial choices are then clustered into two other condition levels as well. Specifically the distance-only and salience-only groups were also analyzed. The strategy groups are shaded and labeled on the y axis policy bands based on the figure 3 policy distribution. The individual dots participants are then moved across the x axis based on the proportion of distance only C4 trials where they chose the closer target, and dot size the proportion of times during reliability only C5 trials where they chose the reliable target. The mean of each C2 cluster's policy values across the C4 and C5 trials were calculated with the mean C4 being visualized on the graph through the boxes.

The cluster center values where: Cost minimizer (C2 = 12%, C4 = 95%, C5 = 64%), Mixed/balanced (C2 = 54%, C4 = 60%, C5 = 59%), Reliability-prioritizer (C2 = 85%, C4 = 68%, C5 = 93%)

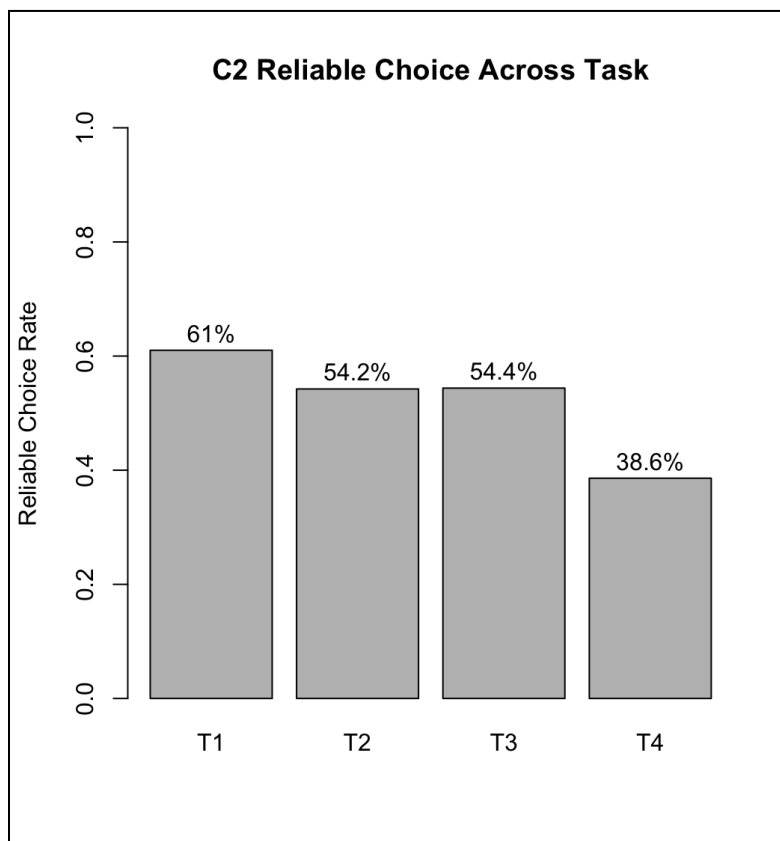


Figure 5. Changes in choice behavior over time. The proportion of reliable choices across four time bins (T1–T4). Results suggest a decrease in reliability preference over time.

Trials were divided into four equal time bins (T1–T4) based on trial number, with each bin containing 10 trials (T1: 1–10, T2: 11–20, T3: 21–30, T4: 31–40). Proportion of selecting the reliable target across time bins (T1–T4) (Figure 5).

Participants chose the reliable target on 61.0%, 54.2%, 54.4%, and 38.6% of trials in T1–T4, suggesting a decrease in reliability preference over time.

A logistic regression analysis showed a significant negative effect of trial number on reliable choice ($\beta = -0.031$, $SE = 0.012$, $z = -2.57$, $p = .010$, $OR = 0.97$, 95% CI [0.95, 0.99]), indicating that participants became less likely to choose the reliable target as the task progressed.

Discussion (Written by: Patrick, Paige)

Overall, our results primarily suggest two things. First, we did not observe strong post-error adjustment in policy. One possible neural interpretation is that although error-related signals are typically associated with regions such as the anterior cingulate cortex and striatum, those signals may not have strongly influenced subsequent value updating or action selection in this task. This could be due to trial feedback signals being too weak (not salient enough or informative to drive behavioral change). Furthermore, these results are consistent with research suggesting that while the prefrontal cortex monitors feedback from recent trials, it may not be responsible for directly driving immediate changes in behavior (Spellman et al., 2021).

But we do show a clear clustering of groups where they prioritize one aspect of the targets which can be explained neurally by how they weigh decision signals more strongly. Reliability-prioritizers may have relied more on parietal evidence and confidence-related signals, while cost-minimizers may have weighted action-cost signals more strongly in cingulate and frontal-striatal valuation systems. Additionally, the differences in how confidence is measured (or processed) may contribute to these decision making strategies, as orbitofrontal cortex neurons have been shown to encode decision confidence and influence later choice behavior (Masset et al., 2020).

The interesting thing about this is that no participant chooses the objective 'best' choice 100% of the time. Specifically based on the cluster groups, participants seemed to correctly identify the optimal choice across one aspect of the target (salience or distance) at the cost of correctly choosing the optimal choice across the other aspect. This behavior may reflect processes that monitor conflicts in the anterior cingulate cortex, which has been shown to detect competition between alternatives and help direct changes in behavior (Botvinick et al., 2001). Additionally, these results are consistent with Fitts' Law, which is the idea that the farther away a target is, the harder it is to process (and the longer it takes to reach) (Fitts, 1954). In this context, closer targets decrease the required movement effort, while more salient targets lessen perceptual uncertainty, which highlights how both motor and perceptual constraints shape human behavior.

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